



Structure and dynamics of differentially rotating plasmas in pulsed-power driven experiments

[Valenzuela-Villaseca+ PRL 2023; JPP (in prep.)]

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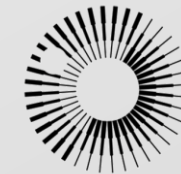
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Driven High-Energy-Density Science



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Supported by NNSA under DOE Cooperative Agreement No DE-SC0020434
and DE-NA0003764, and the Imperial College President's PhD Scholarships.

Rotating plasmas are formed in the universe when an accretion disk orbits a central object

Angular Momentum Transport Problem

$$\tau_{\text{acc,visc}} \sim 10^3 \tau_{\text{acc,obs}}$$

What mechanism removes angular momentum from the rotating flow, enhancing the accretion rate?

Standard theory of accretion disks

[Balbus 1998]

Differential
rotation
+
Magnetic
fields

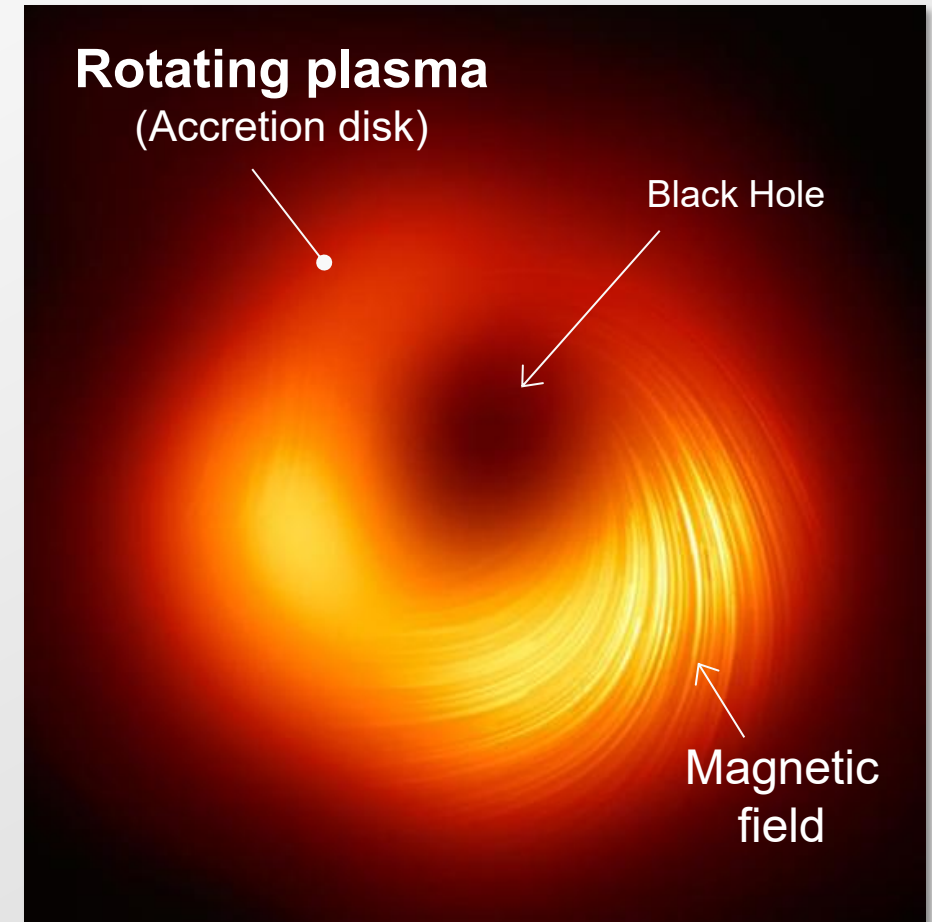


**Magnetorotational
instability**
(Balbus & Hawley 1991)
+
Parasitic instabilities
(Goodman & Xu 1994)



**Anomalous
turbulent viscosity
and enhanced
accretion rate**
(Shakura & Sunyaev 1973)

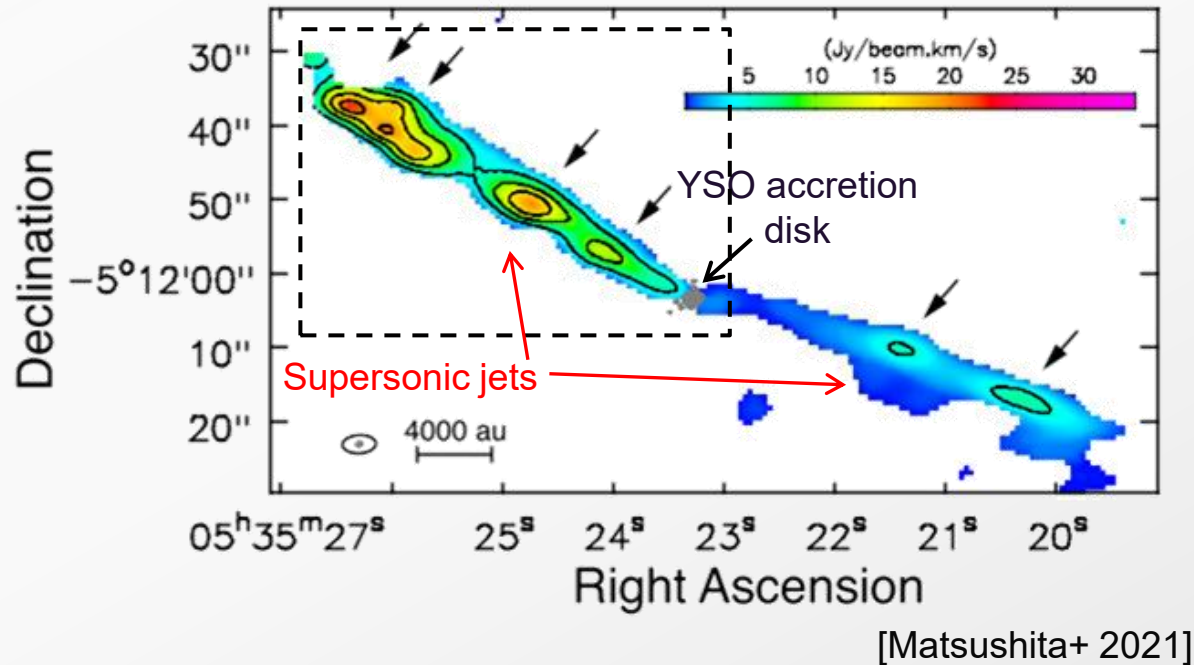
M87 supermassive black hole



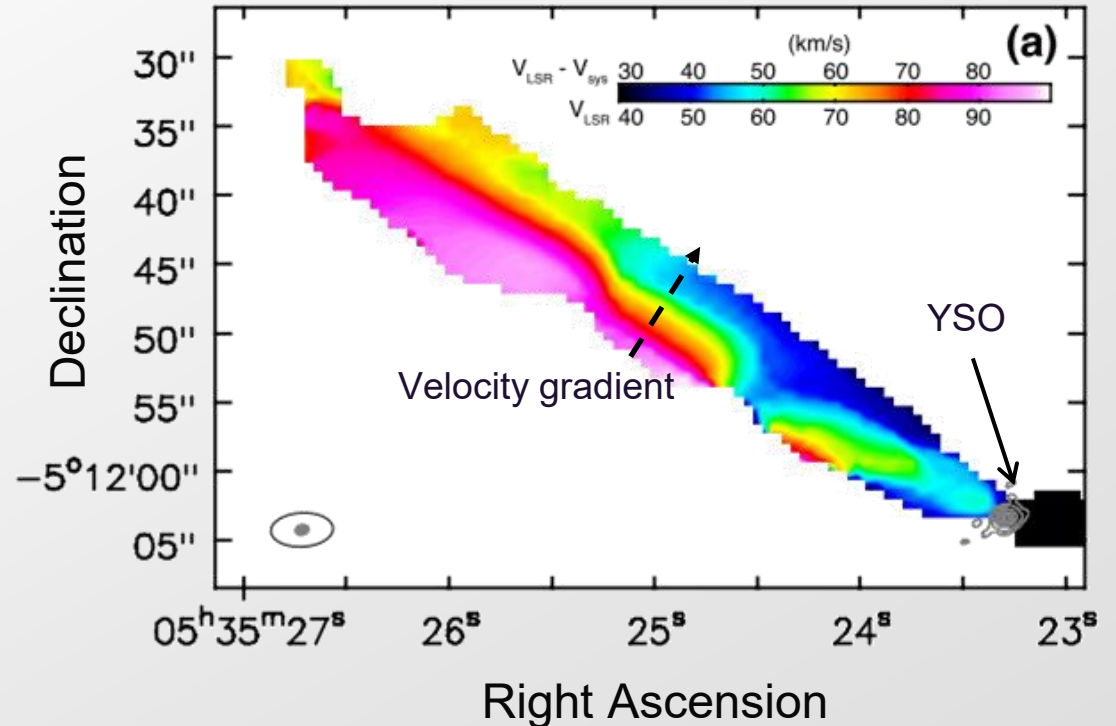
[Akiyama+ (EHT Collab.) ApJ 2021]

Observations of radio jets launched by young stellar objects (YSO) indicate fast rotation and angular momentum transport

FIR 6b, Orion Molecular Cloud (ALMA)



Doppler velocity from CO lines
J=2 – 1, 230.539 GHz



- Velocity gradient across jet indicates rotation ~ 20 km / s
- **How much angular momentum gets evacuated by these jets?**

Rotation velocity stratification is a key characteristic of accretion disks which defines their stability

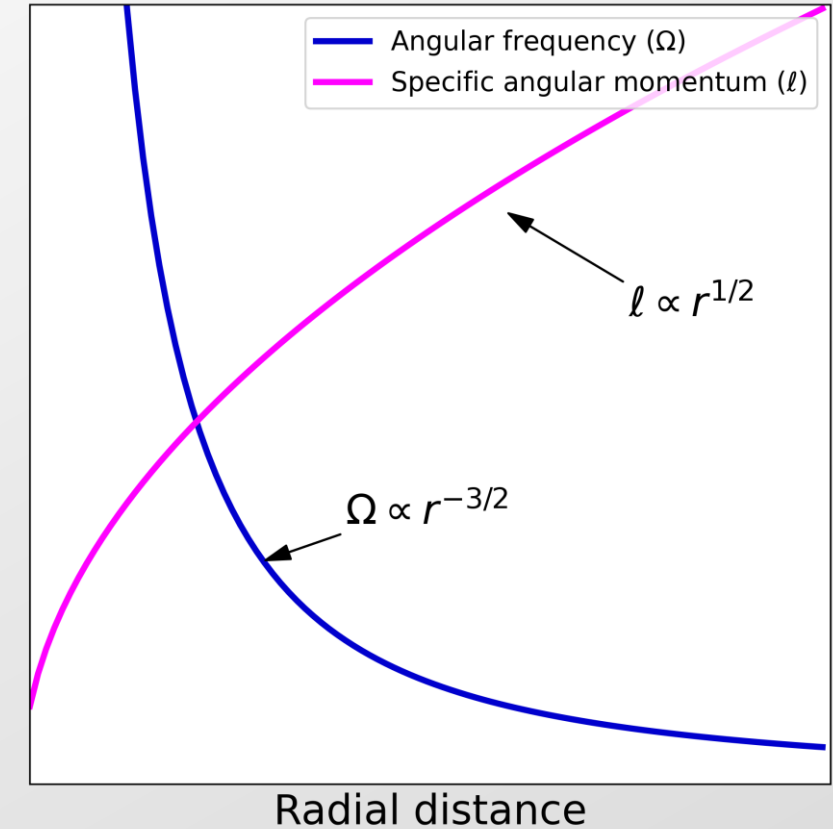
Essential MHD description of the accretion disk

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times \mathbf{u} \times \mathbf{B} + \eta \nabla^2 \mathbf{B}$$

$$\frac{d}{dt} \rho \mathbf{u} = -\nabla(p + p_M + \Phi) + \frac{1}{\mu_0} (\mathbf{B} \cdot \nabla) \mathbf{B} + \nu \nabla^2 \mathbf{u}$$

Gravitational potential drives orbital motion

$$\mathbf{u}_\theta \propto \mathbf{r}^{-1/2}$$



Rotation velocity stratification is a key characteristic of accretion disks which defines their stability

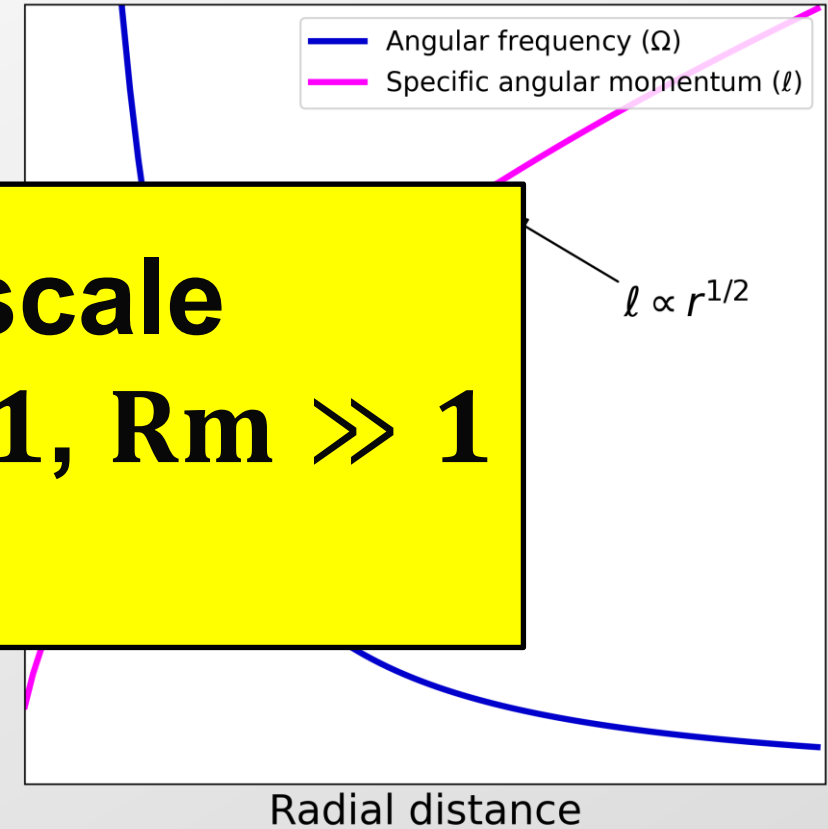
Essential MHD description of the accretion disk

$$\frac{d}{dt} \rho \mathbf{u} = -$$

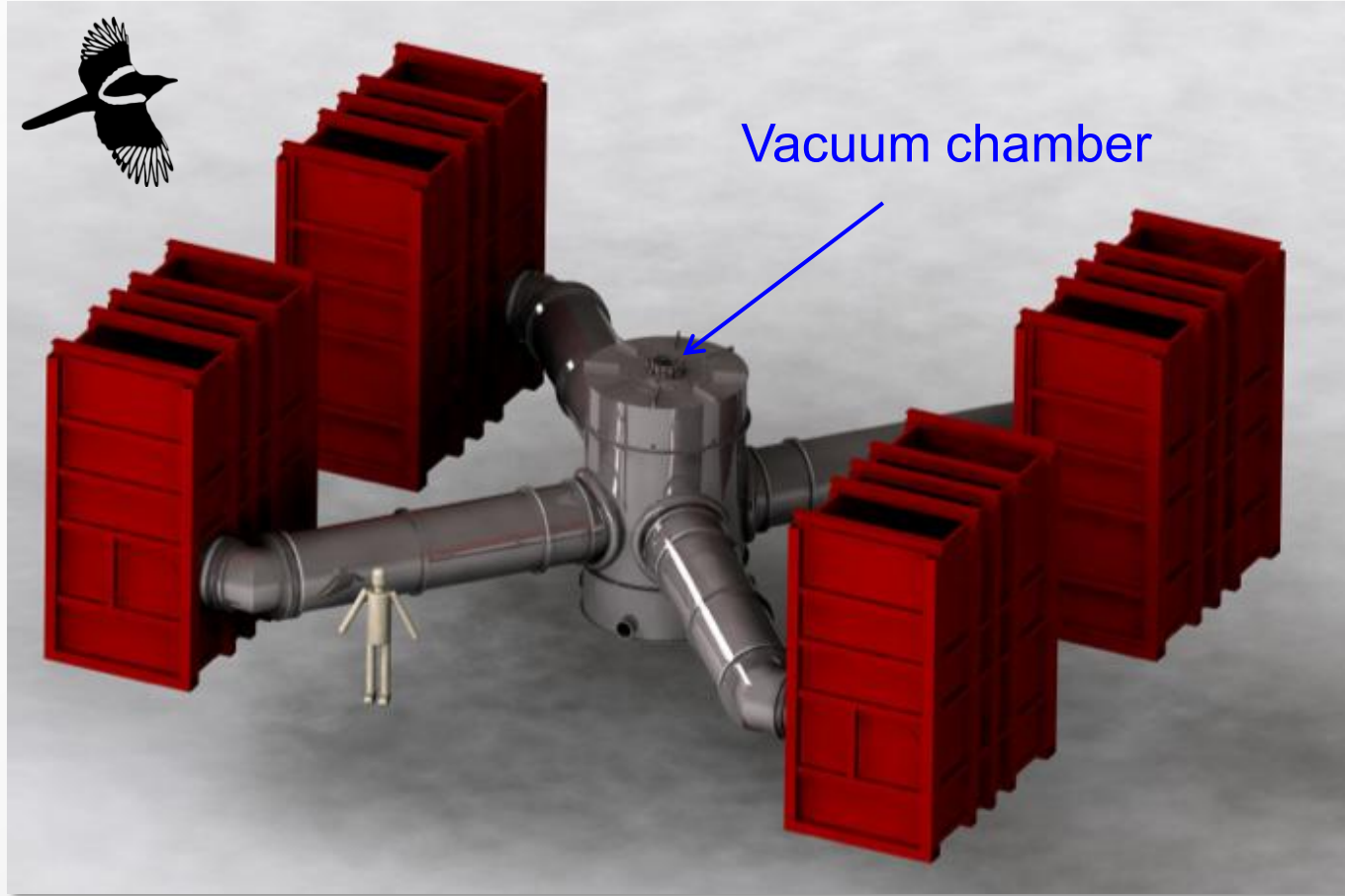
In addition to MHD scale invariance criteria: $Re \gg 1, Rm \gg 1$ (Ryutov+ 1999,...)

Quasi-Keplerian stratification:

1. **Hydrodynamically stable:** $d\ell/dr > 0$ (Rayleigh criterion)
2. **MHD unstable:** $d\Omega/dr < 0$ (Magnetorotational instability)



Rotating plasma experiments were conducted on the MAGPIE pulsed-power generator

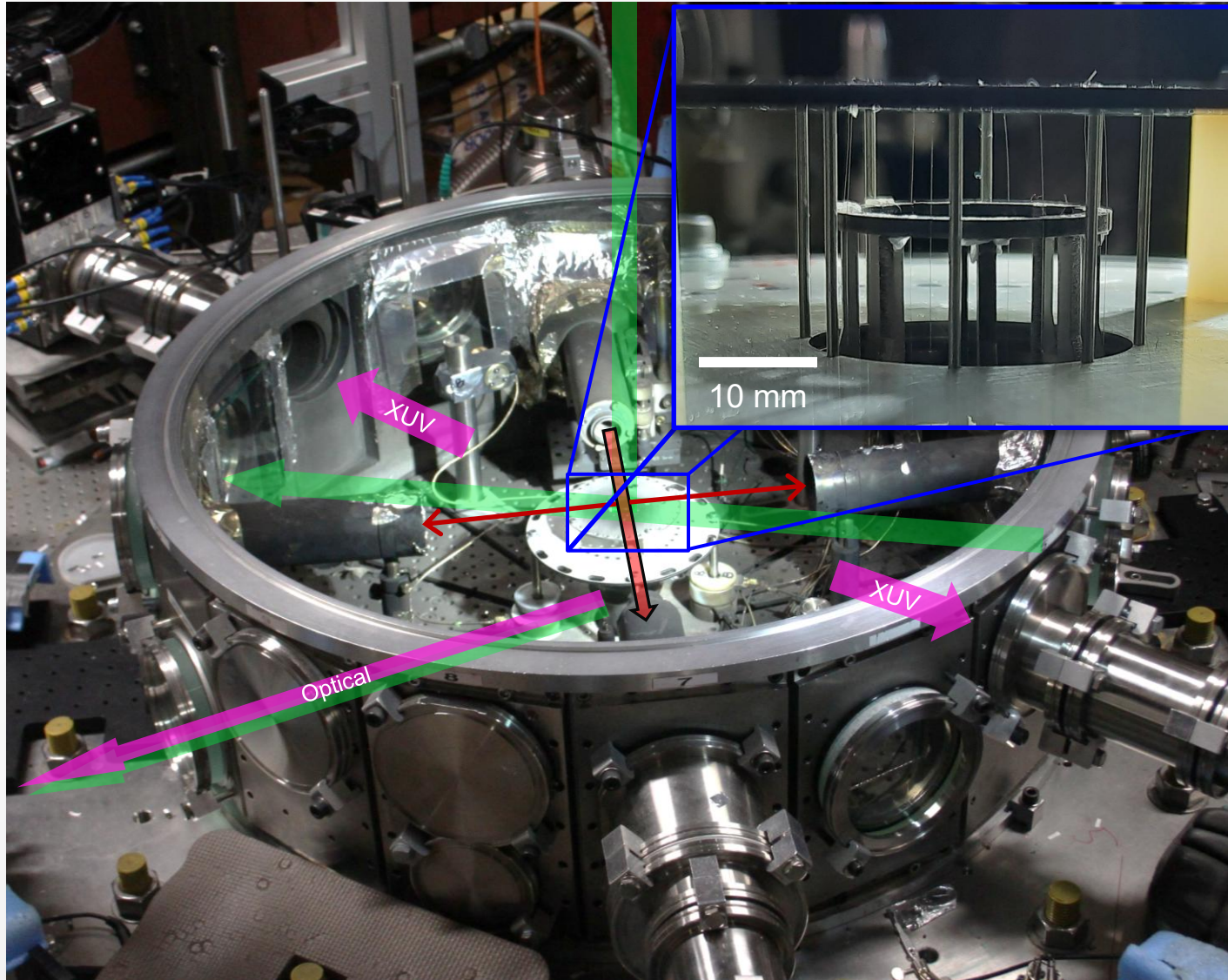


- 1.4 MA peak electrical current
- 240 ns rise-time

[Mitchell+ RSI 1996]

A diagnostic suite allows obtaining measuring multiple plasma parameters on each experiment

[Swadling+ 2014; Suttle+ 2021]



- **Laser imaging and interferometry**
→ density structure

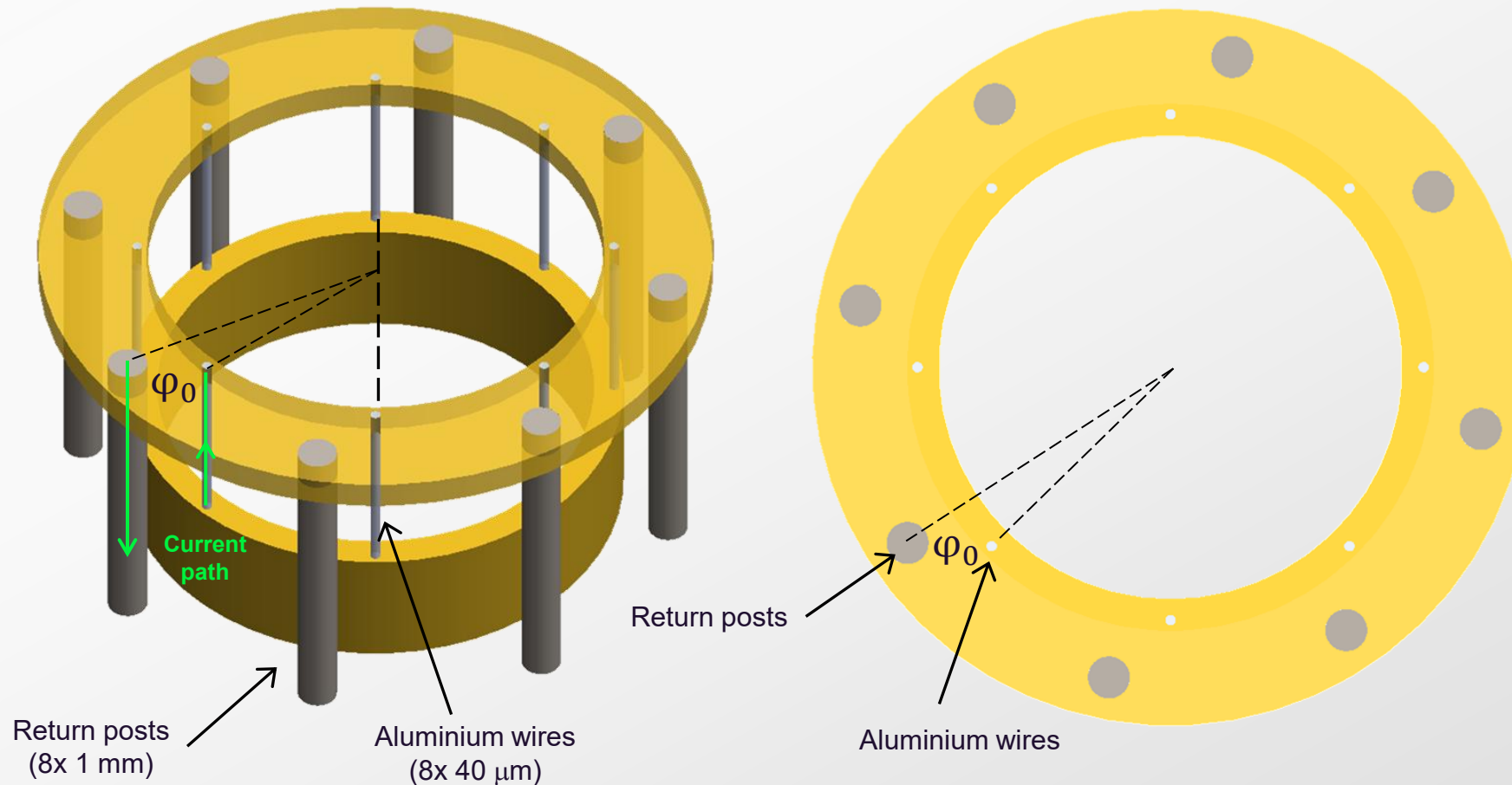
- **Self-emission imaging (optical, XUV)**
→ structure and global dynamics

- **Optical Thomson Scattering**
→ local velocity and temperature

The Rotating Plasma Experiment (RPX) drives rotation on a wire array Z-pinch plasma

3-D view

Vertical view

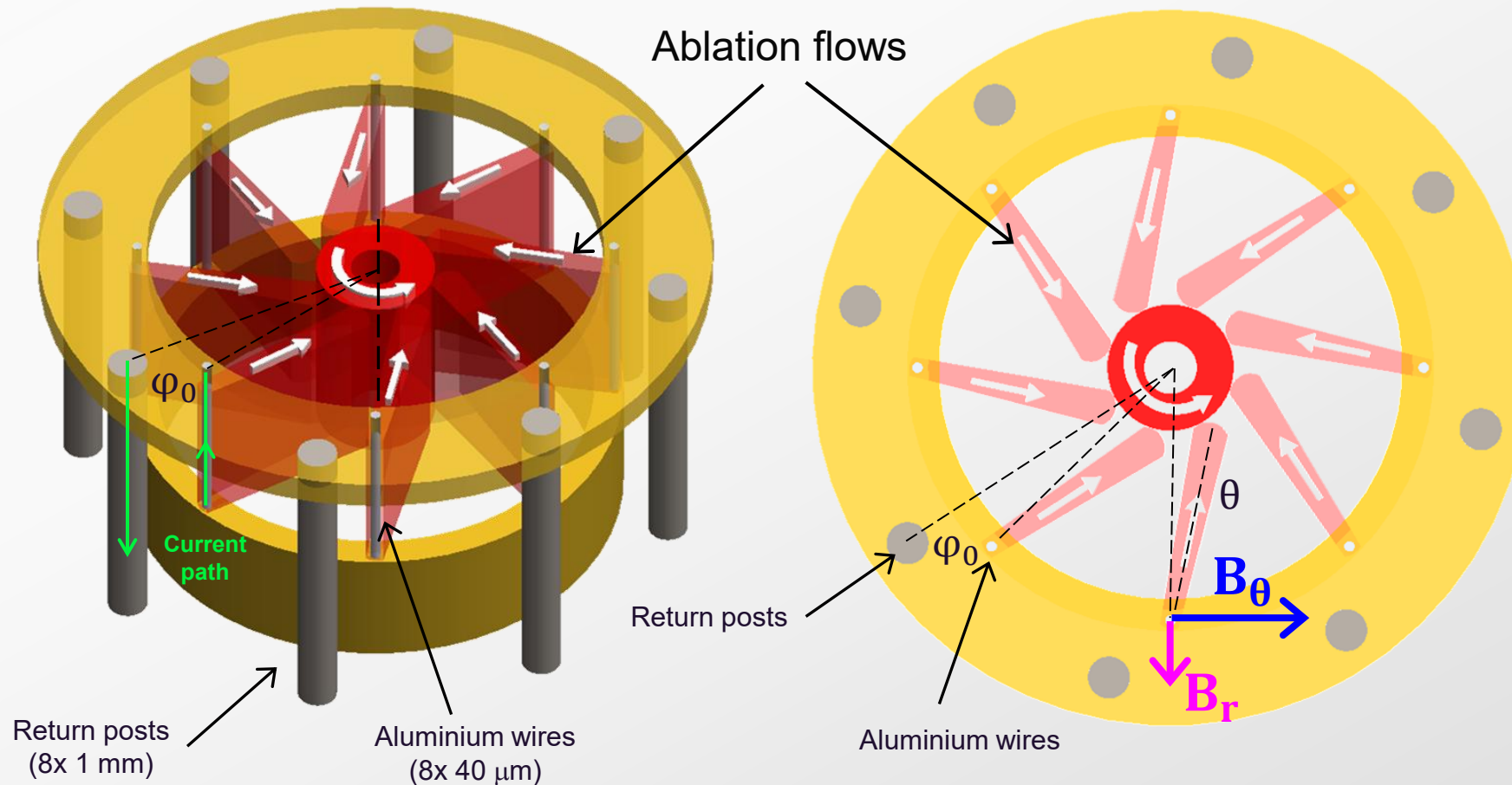


[Valenzuela-Villaseca+ PRL 2023]

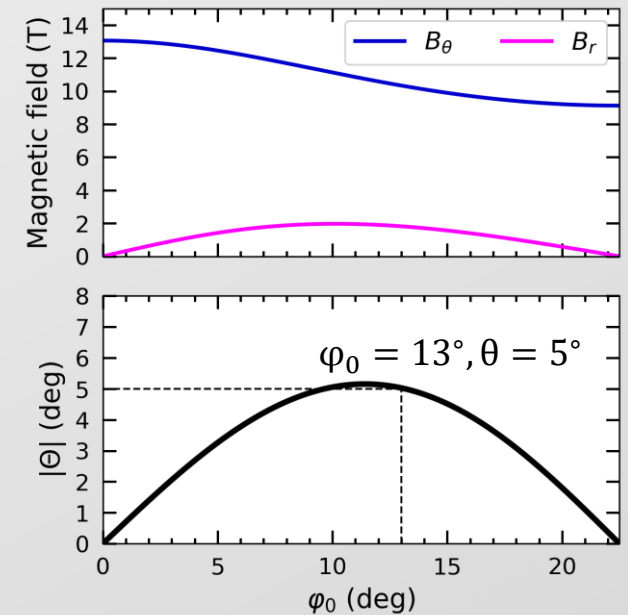
The Rotating Plasma Experiment (RPX) drives rotation on a wire array Z-pinch plasma

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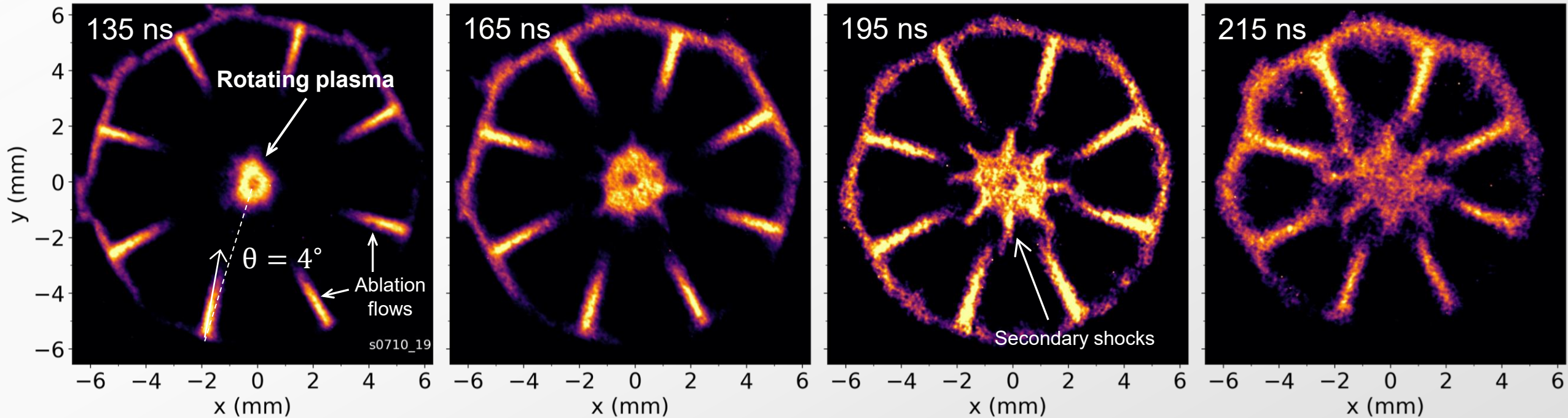
1. RPX configuration produces B_r and B_θ components
2. $\mathbf{J} \times \mathbf{B}$ force has a radial azimuthal components
3. Ablation flows are accelerated azimuthally and radially inwards, parametrized by θ
4. They collide and produce a **rotating plasma on-axis**



[Valenzuela-Villaseca+ PRL 2023]

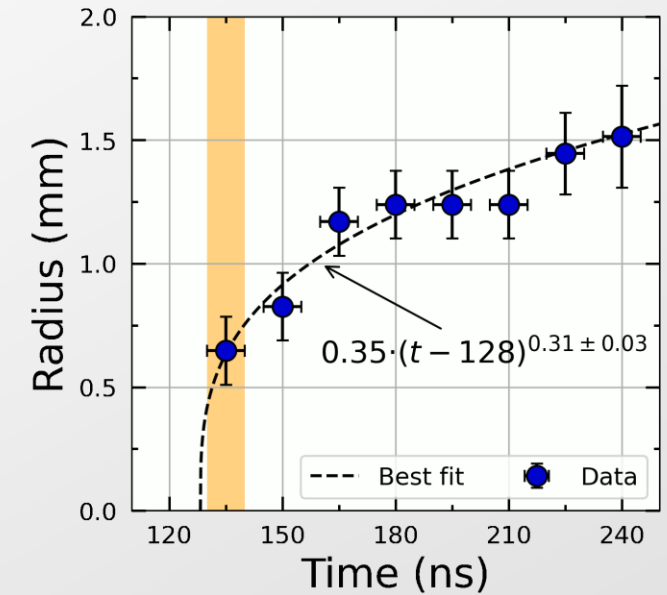
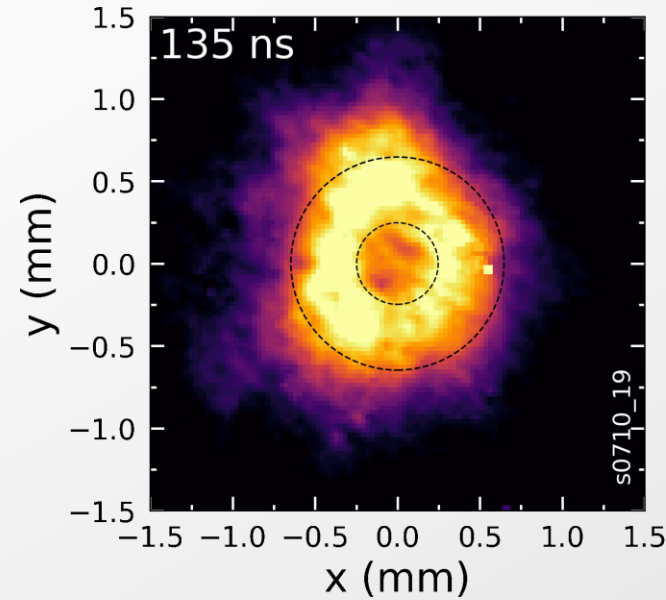
Dynamically significant angular momentum sustains hollow rotating plasmas

Vertical view – optical self-emission imaging



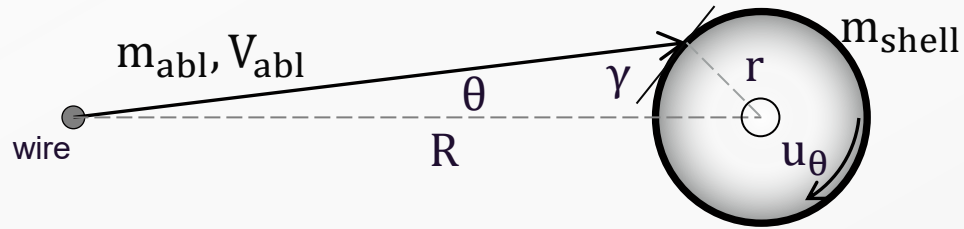
- Measured θ is consistent with analytical estimation
- **Hollow rotating plasma** forms on the axis
- Plasma accumulates on a stagnation surface
- Centrifugal barrier: $\frac{d}{dr}(\rho_{ab} u_r^2) = \rho \frac{u_\theta^2}{r}$
- **Secondary shocks** form due to oblique collision of ablation flows
- Rotating plasma grows radially due to mass accumulation

Time evolution of rotating plasmas can be described by simple conservation laws



Time evolution of rotating plasmas can be described by simple conservation laws

- Using a similar setup to [Ampleford+ PRL 2008]



- Mass conservation

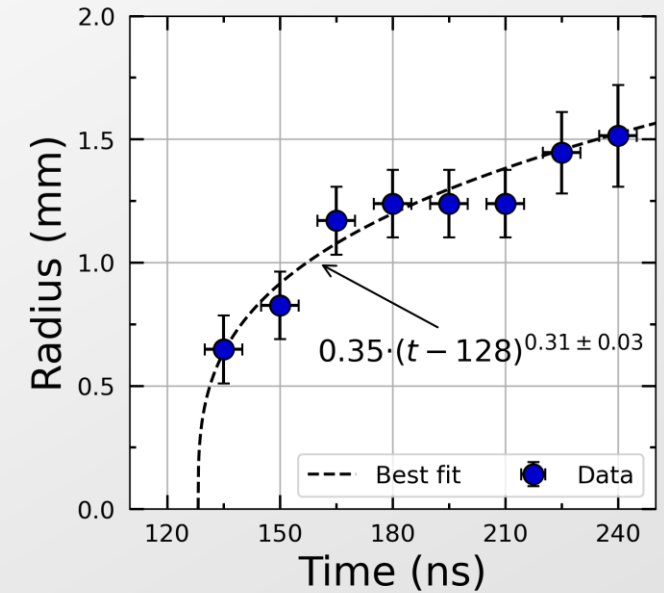
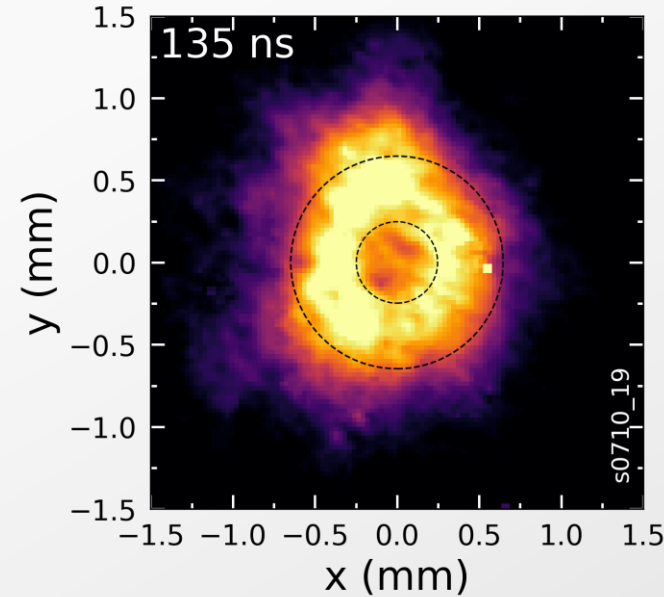
$$m_{\text{shell}} \approx \frac{dm_{\text{abl}}}{dt} (t - t_{\text{ret}}) \sim \frac{dm_{\text{abl}}}{dt} \left(t - R/V_{\text{abl}} \right)$$

- Conservation of angular momentum

$$R V_{\text{abl}} \sin(\theta) = r u_{\theta}$$

- Conservation of momentum

$$\frac{m_{\text{shell}} u_{\theta}^2}{r} = V_{\text{abl}} \sin(\gamma) \frac{dm_{\text{abl}}}{dt}$$

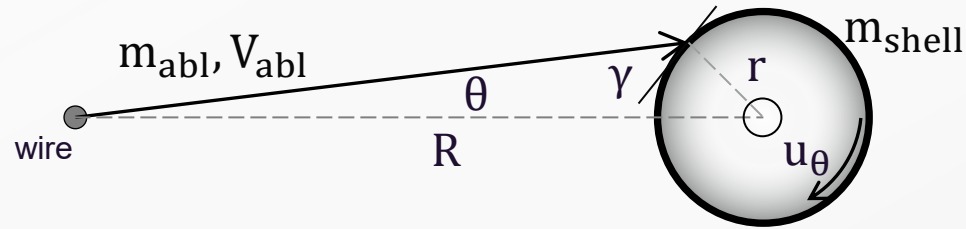


$$\rightarrow r(t) = \left[\frac{\sin^2(\theta)}{\sin(\gamma)} R^2 V_{\text{abl}} (t - R/V_{\text{abl}}) \right]^{1/3} \sim t^{1/3}$$

Confinement provided by the ablation flows' ram pressure,
not the magnetic field

Time evolution of rotating plasmas can be described by simple conservation laws

- Using a similar setup to [Ampleford+ PRL 2008]



- Mass conservation

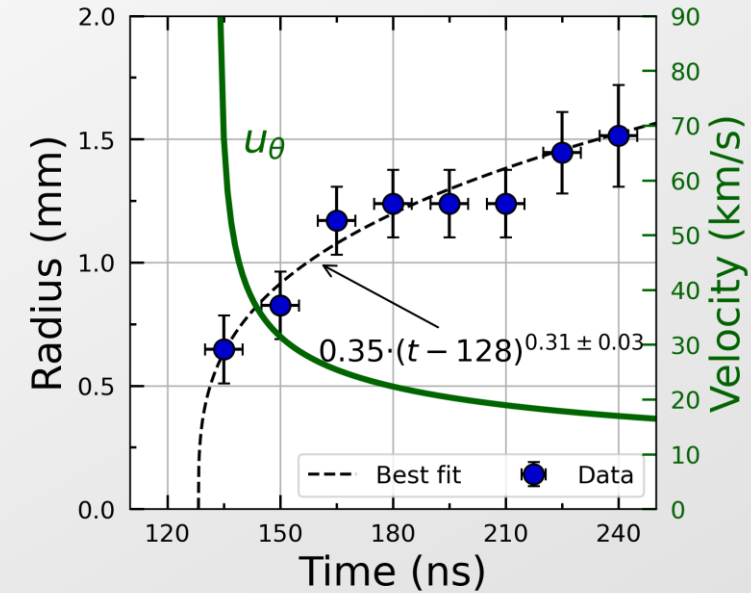
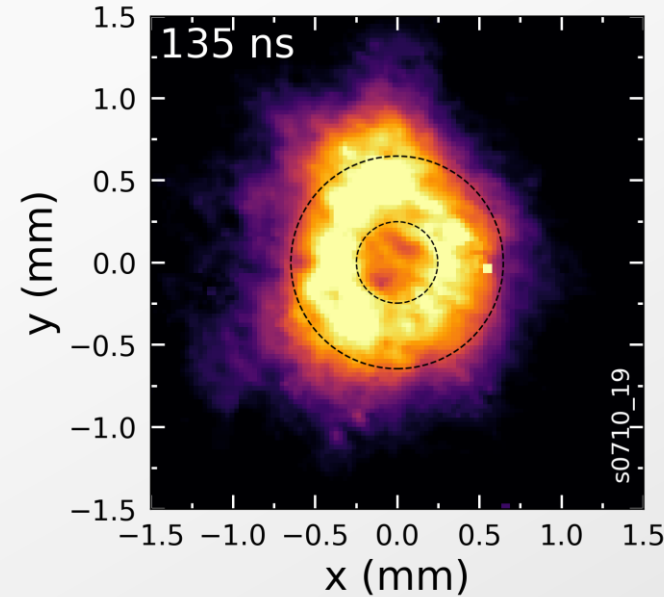
$$m_{\text{shell}} \approx \frac{dm_{\text{abl}}}{dt} (t - t_{\text{ret}}) \sim \frac{dm_{\text{abl}}}{dt} \left(t - R/V_{\text{abl}} \right)$$

- Conservation of angular momentum

$$RV_{\text{abl}} \sin(\theta) = ru_{\theta}$$

- Conservation of momentum

$$\frac{m_{\text{shell}} u_{\theta}^2}{r} = V_{\text{abl}} \sin(\gamma) \frac{dm_{\text{abl}}}{dt}$$

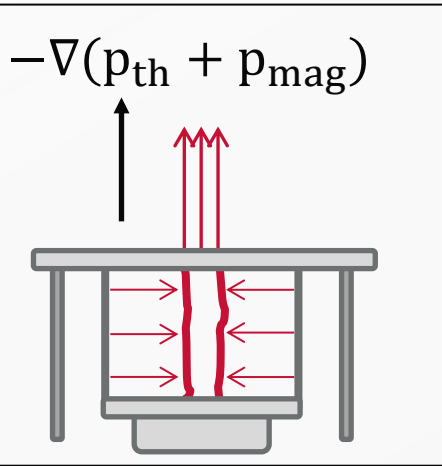


$$\rightarrow r(t) = \left[\frac{\sin^2(\theta)}{\sin(\gamma)} R^2 V_{\text{abl}} (t - R/V_{\text{abl}}) \right]^{1/3} \sim t^{1/3}$$

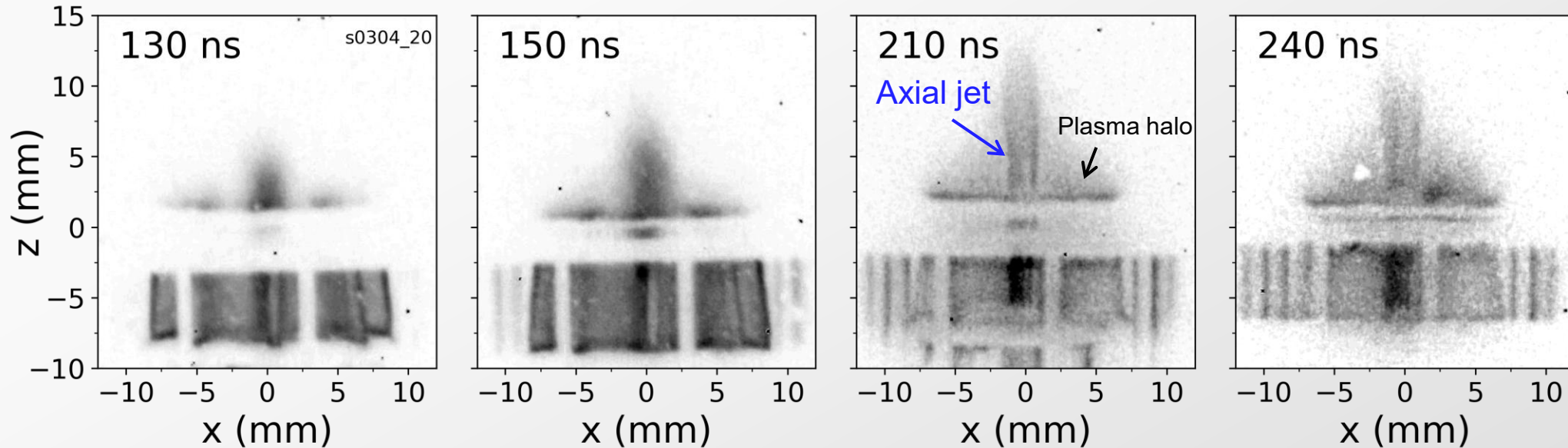
$$\rightarrow u_{\theta}(t) = \left[\sin(\theta) \sin(\gamma) \frac{RV_{\text{abl}}^2}{t - R/V_{\text{abl}}} \right]^{1/3} \sim 15 - 20 \text{ km/s}$$

Axial jets are launched outwards by pressure gradients

Schematic view



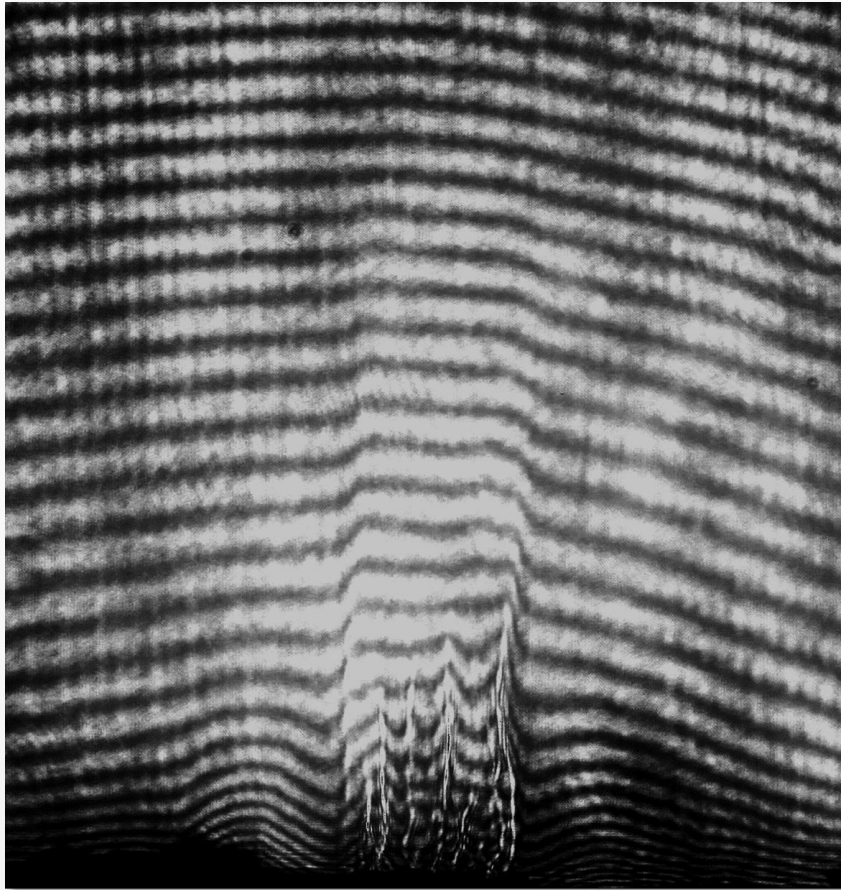
Lateral view – XUV imaging



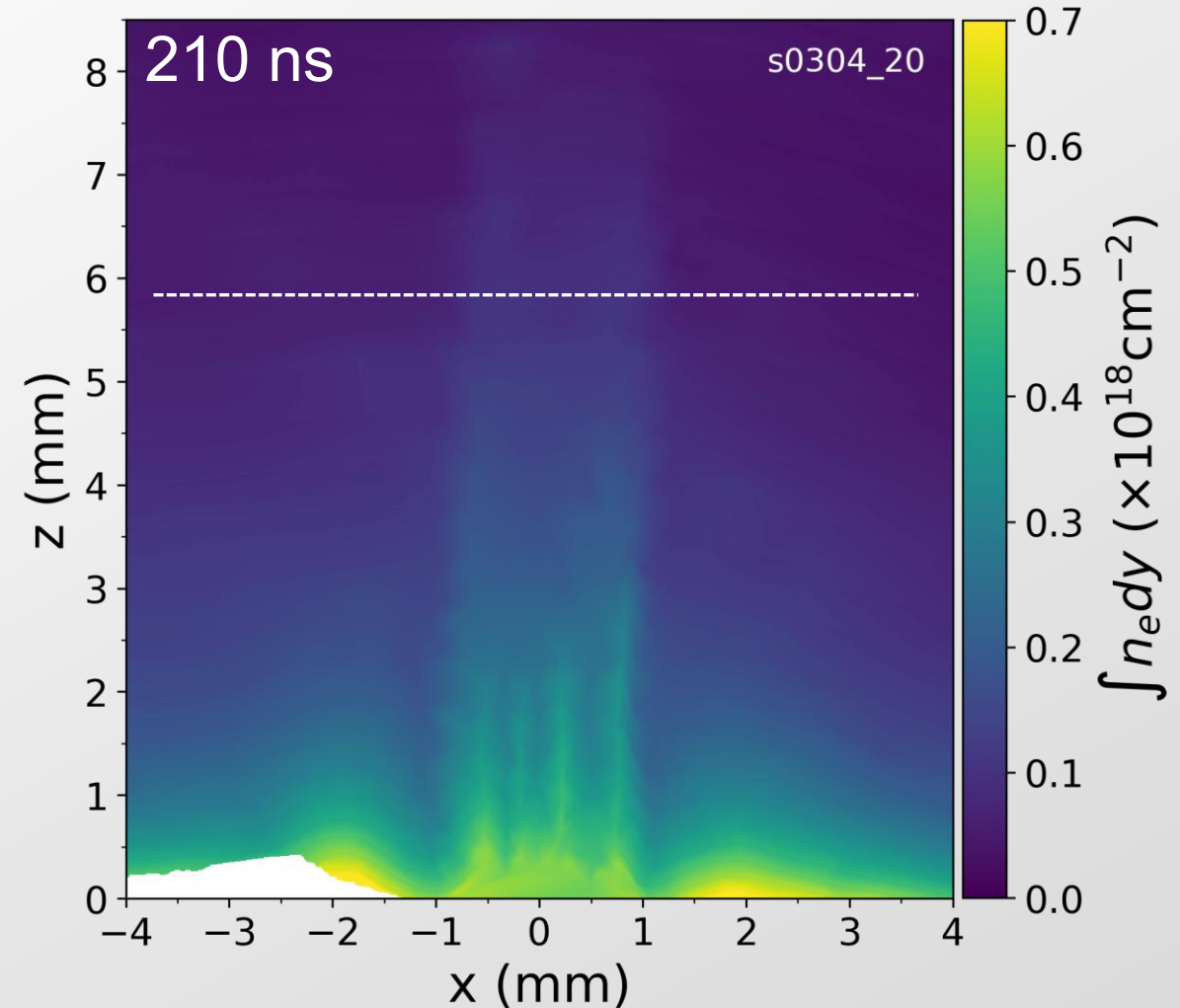
- Axial velocity ~ 100 km/s
- Divergence angle: ~ 3 degrees
- Collimation provided by plasma halo
- No significant change in morphology until ~ 270 ns (i.e. ~ 150 ns of steady state)
- Axial jet is perfect for laser diagnostics!

Laser interferometry is used to probe the electron density, showing the jet's on-axis density depletion

Raw laser interferogram



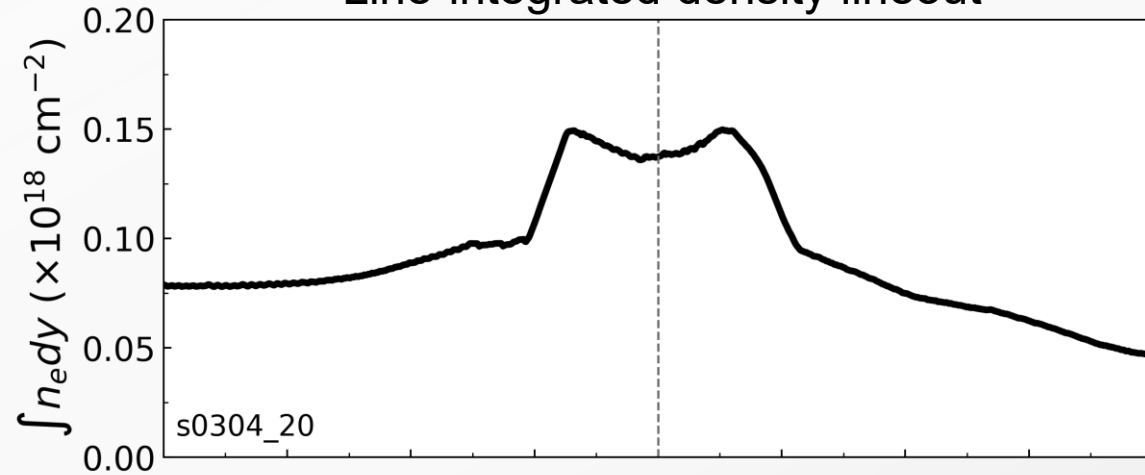
Line-integrated density



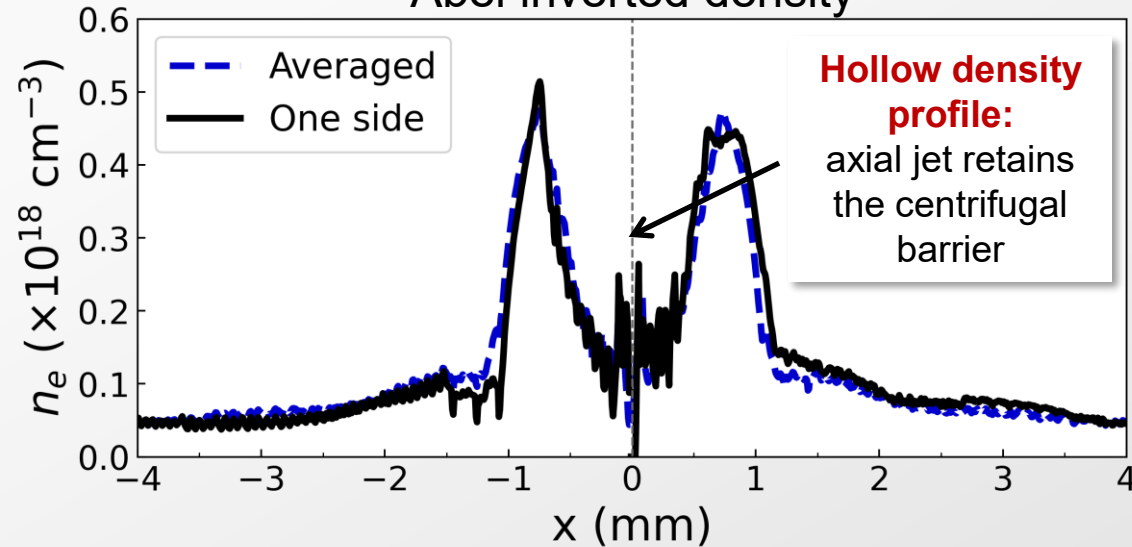
[Valenzuela-Villaseca+ PRL 2023]

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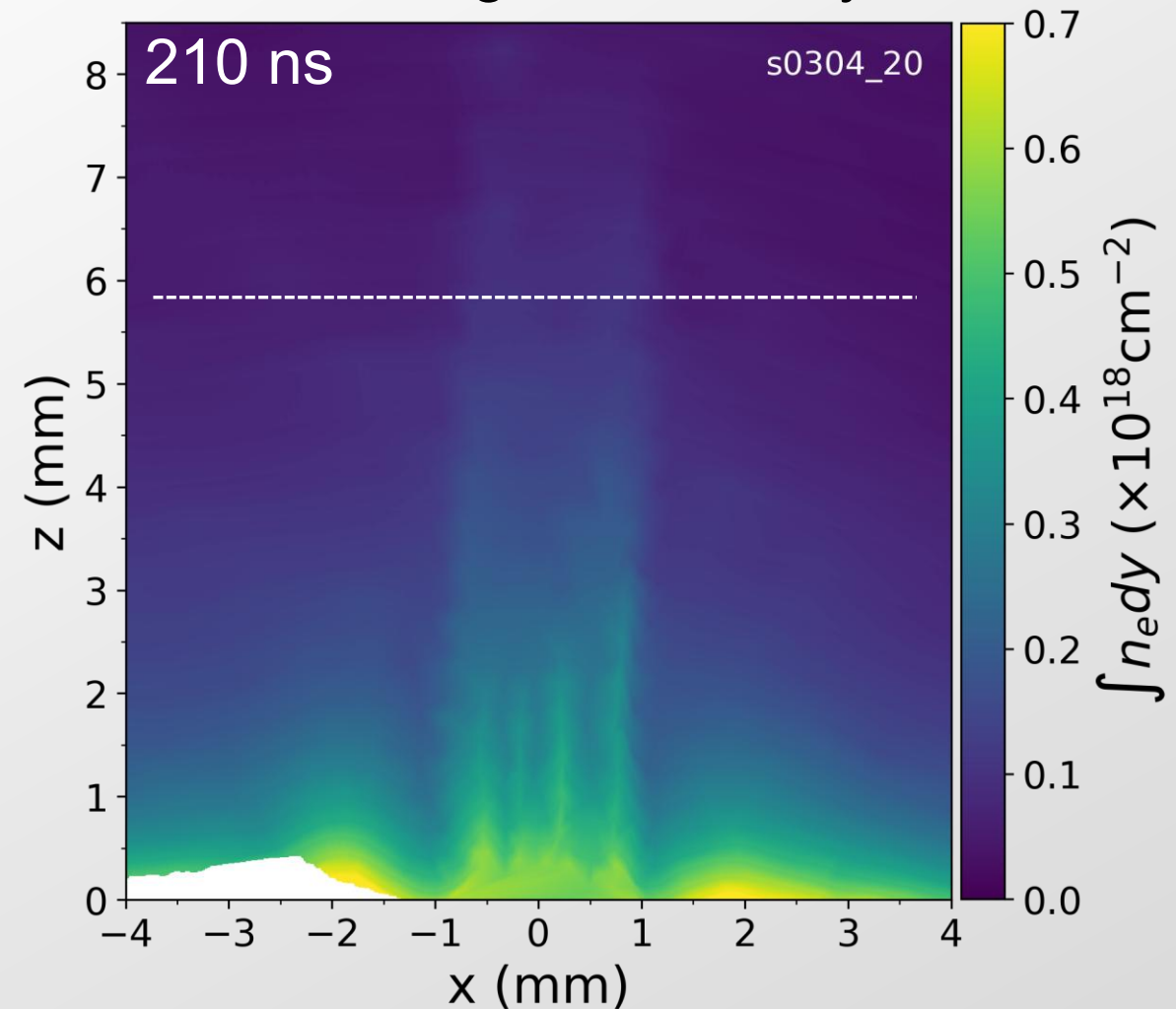
Line-integrated density lineout



Abel inverted density



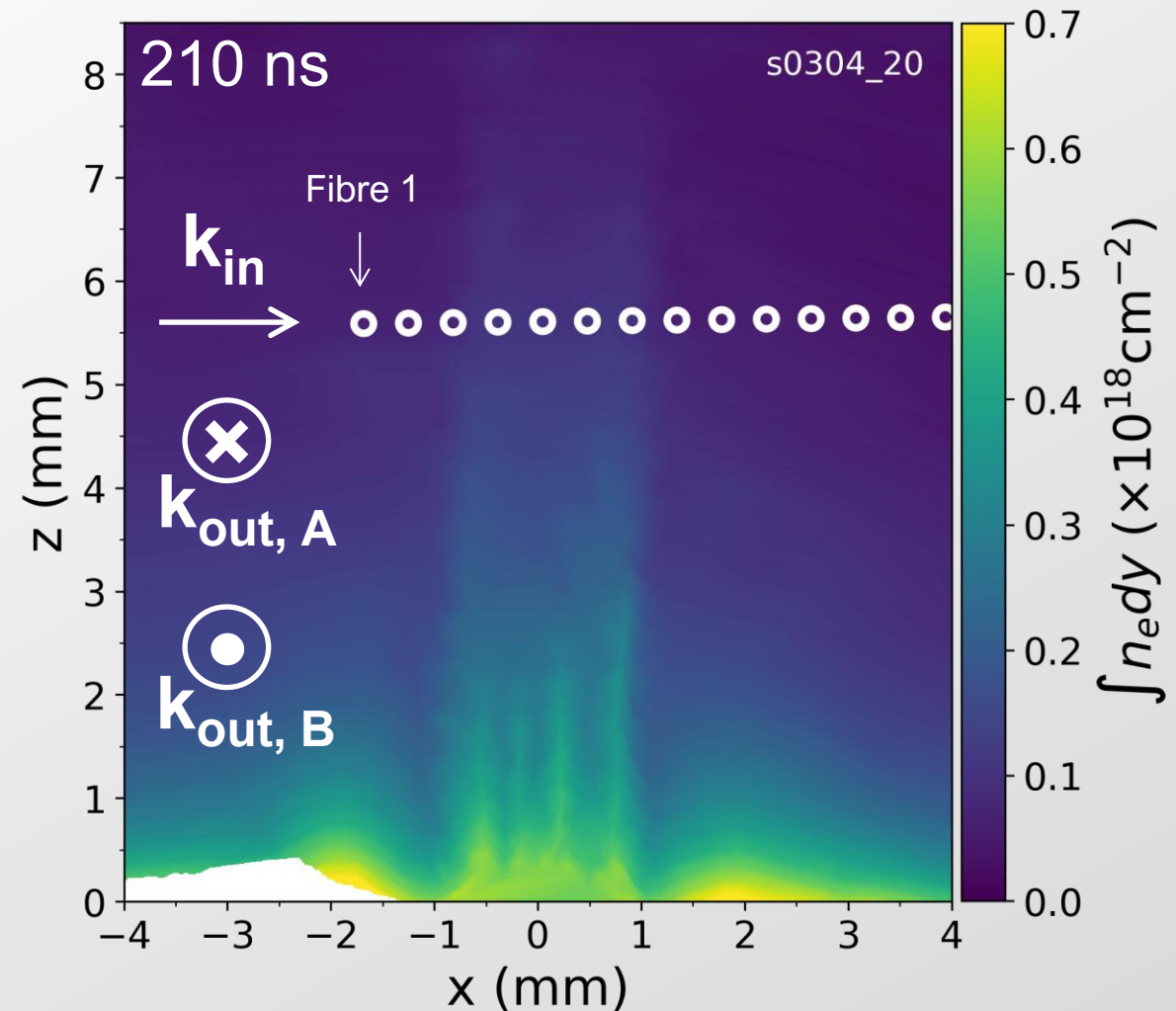
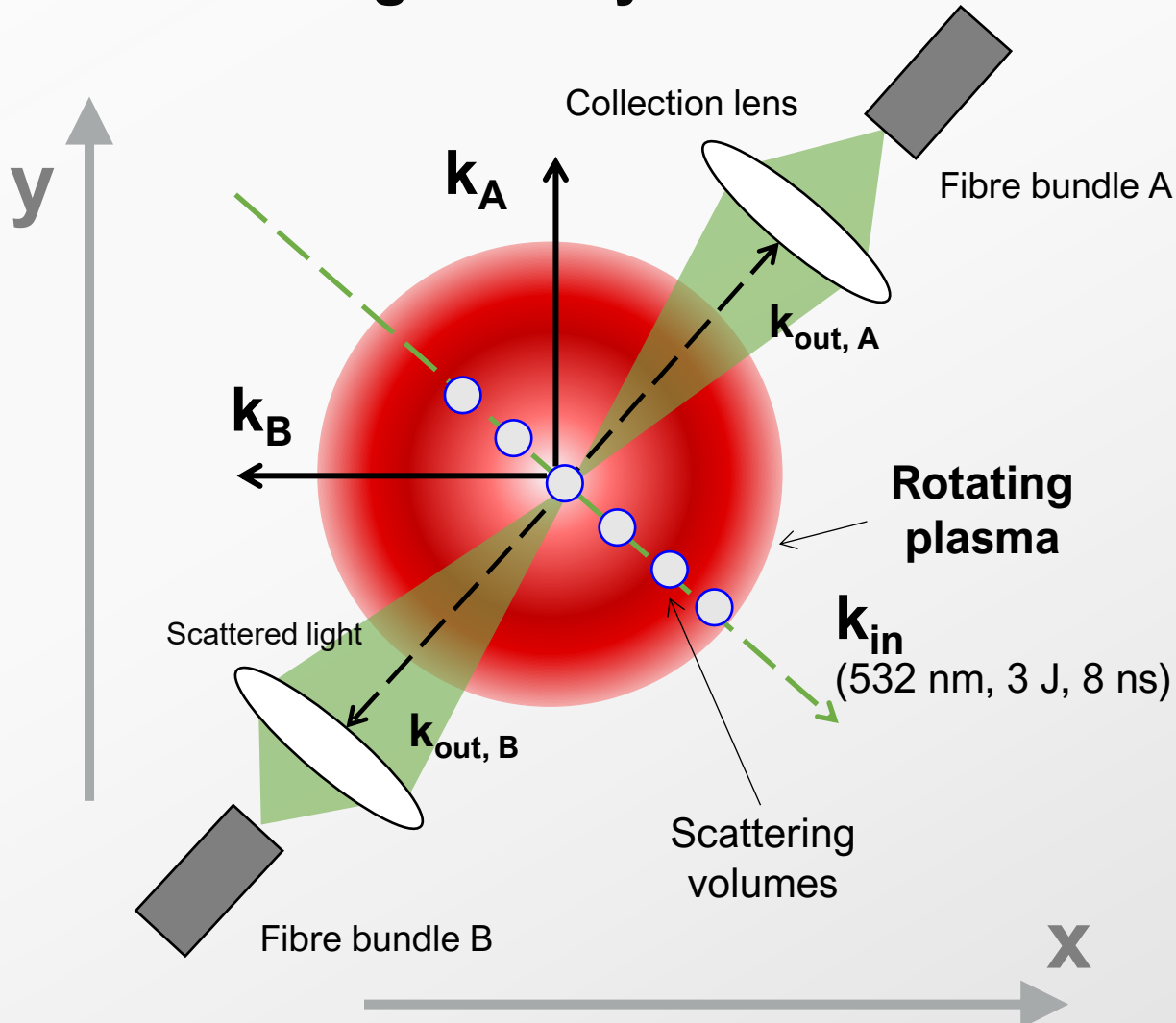
Line-integrated density



[Valenzuela-Villaseca+ PRL 2023]

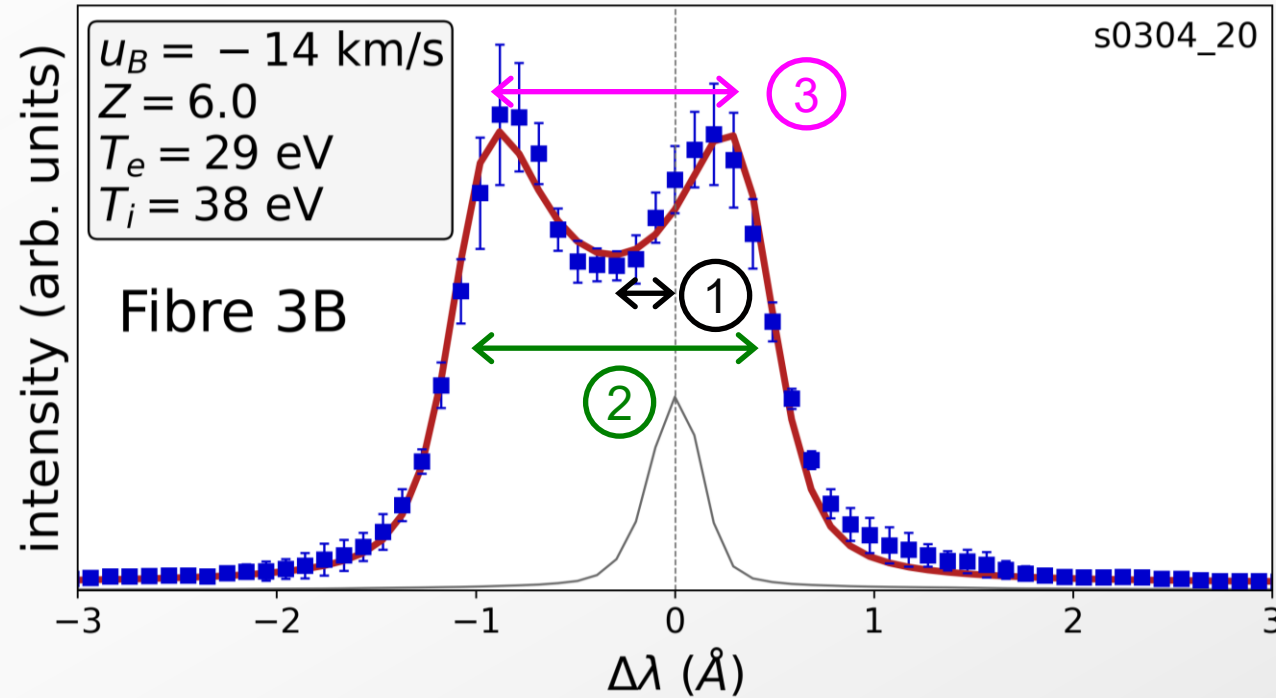
Plasma parameters on the axial jet are probed using an IAW Thomson Scattering diagnostic

TS geometry – end on



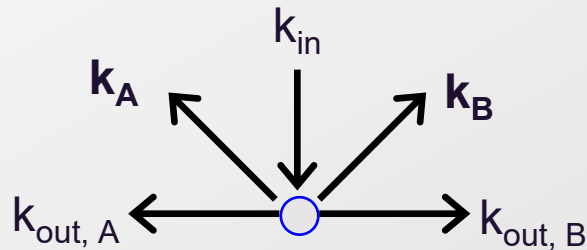
[Valenzuela-Villaseca+ PRL 2023]

IAW spectral features allows obtaining velocity and temperature



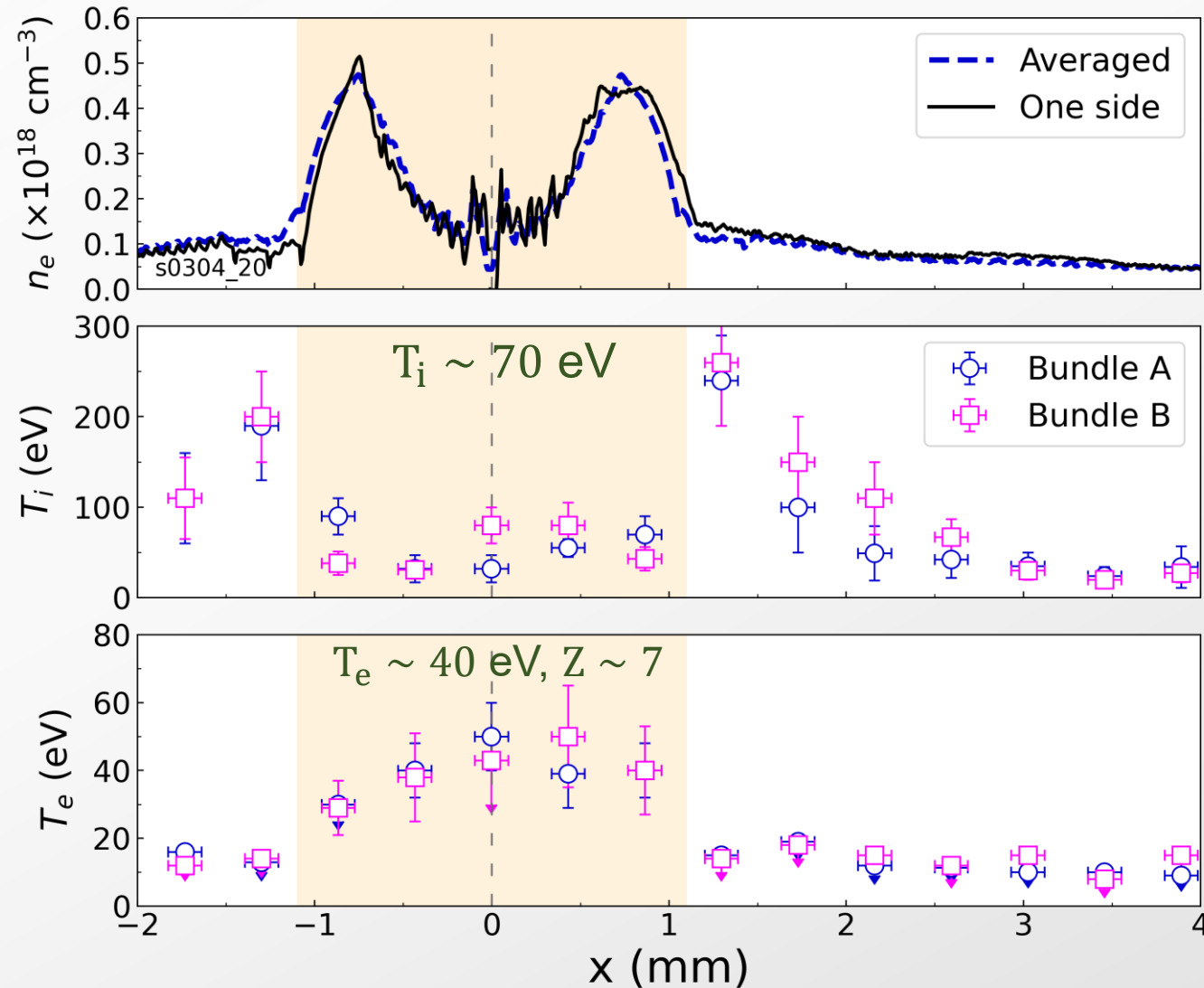
- ① Doppler shift: plasma velocity
- ② Feature width: T_i
- ③ Ion Acoustic peak separation: ZT_e

TS geometry



[Suttle+ RSI 2021, Hare PhD Thesis 2017, Niasse PhD Thesis 2011]

The rotating plasma is well described by near-ideal, single fluid MHD

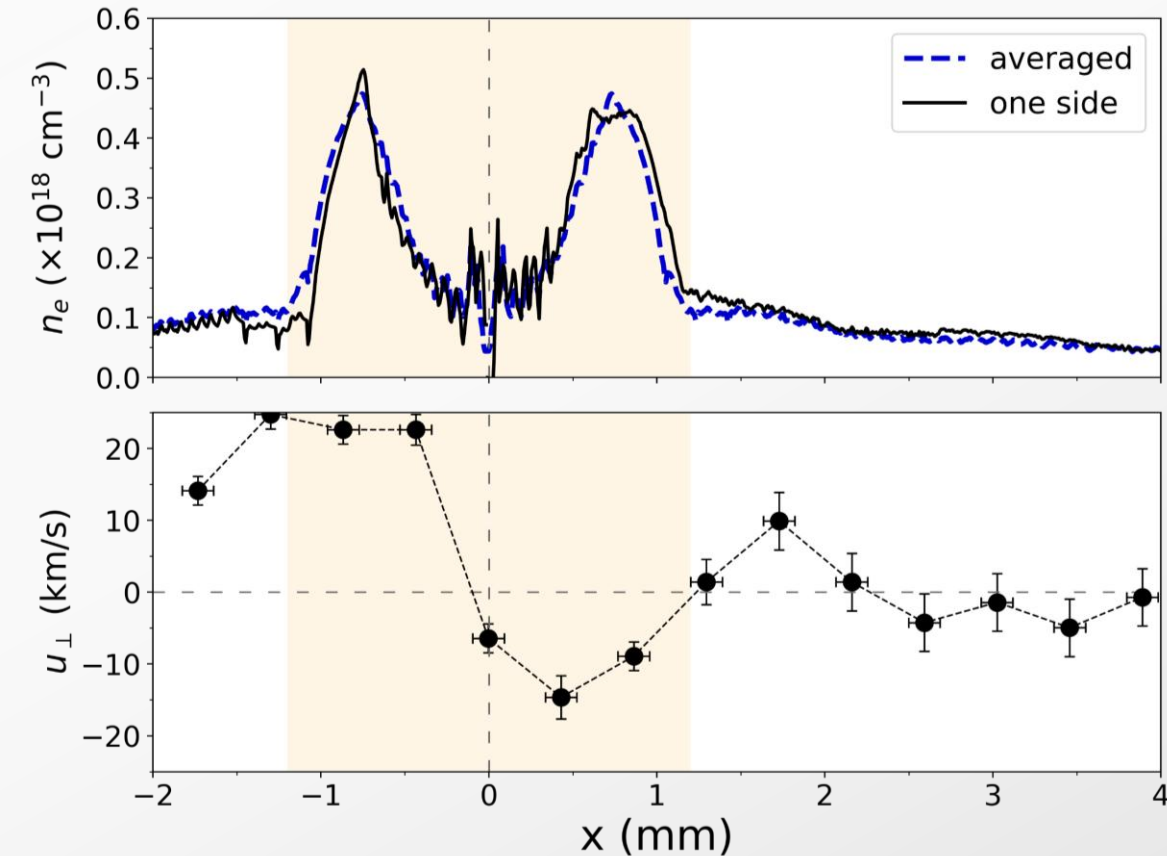


Relevant length scales

- Height $> 15 \text{ mm}$
- Diameter $> 2 \text{ mm}$
- ion-ion mfp $\sim 1 \mu\text{m}$
- $\ell_v \sim 0.5 \mu\text{m}$, $\ell_\eta \sim 70 \mu\text{m}$ ($\text{Pm} \sim 10^{-2}$),
i.e. $\text{Re} \gg 1$, $\text{Rm} \gg 1$, $\text{Re} \gg \text{Rm}$
- $d_i = c/\omega_{pi} \sim 50 \mu\text{m}$

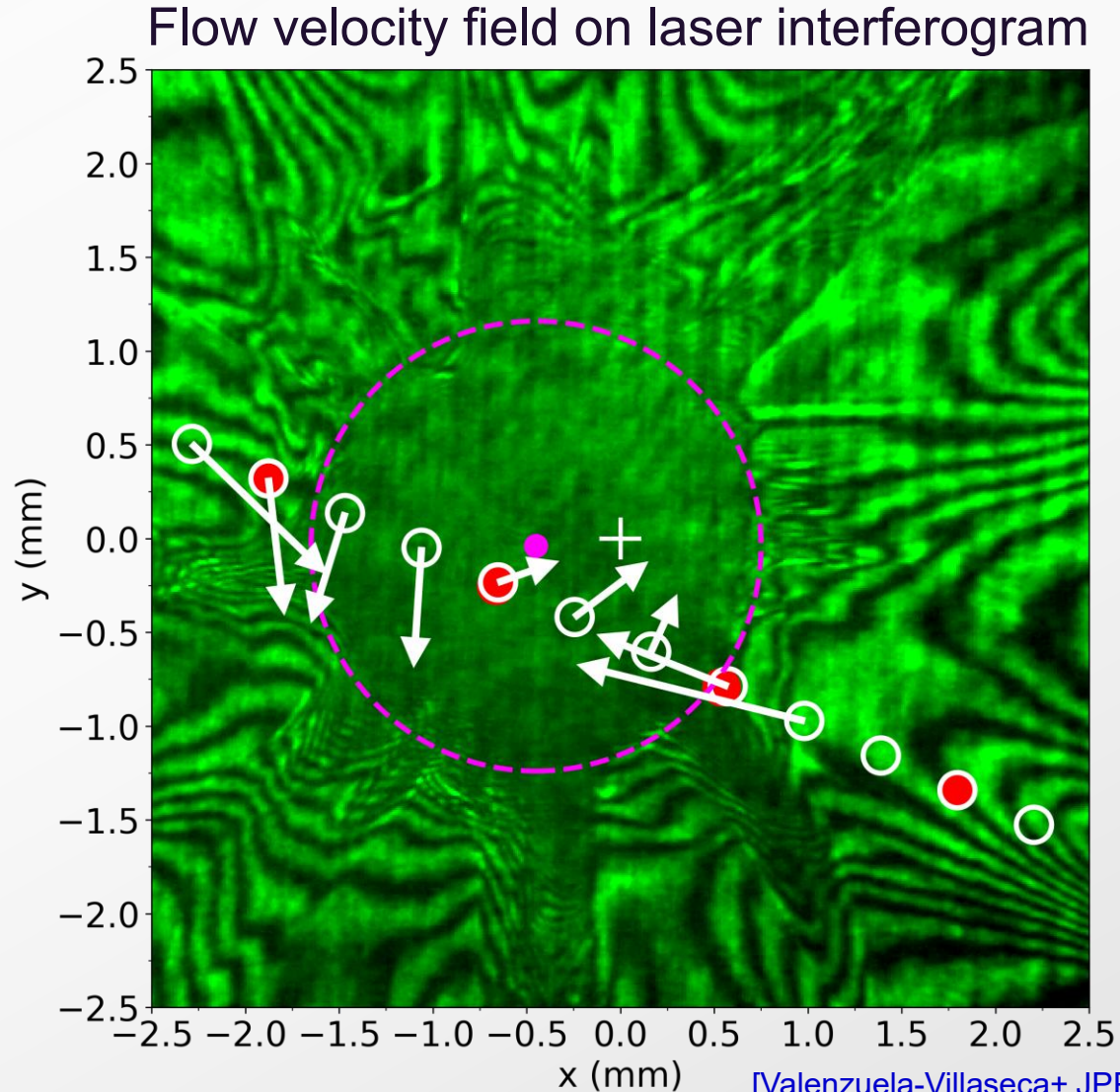
Collisional
Near-ideal, low-Pm
Single fluid MHD plasma

The axial jet rotates and transports angular momentum

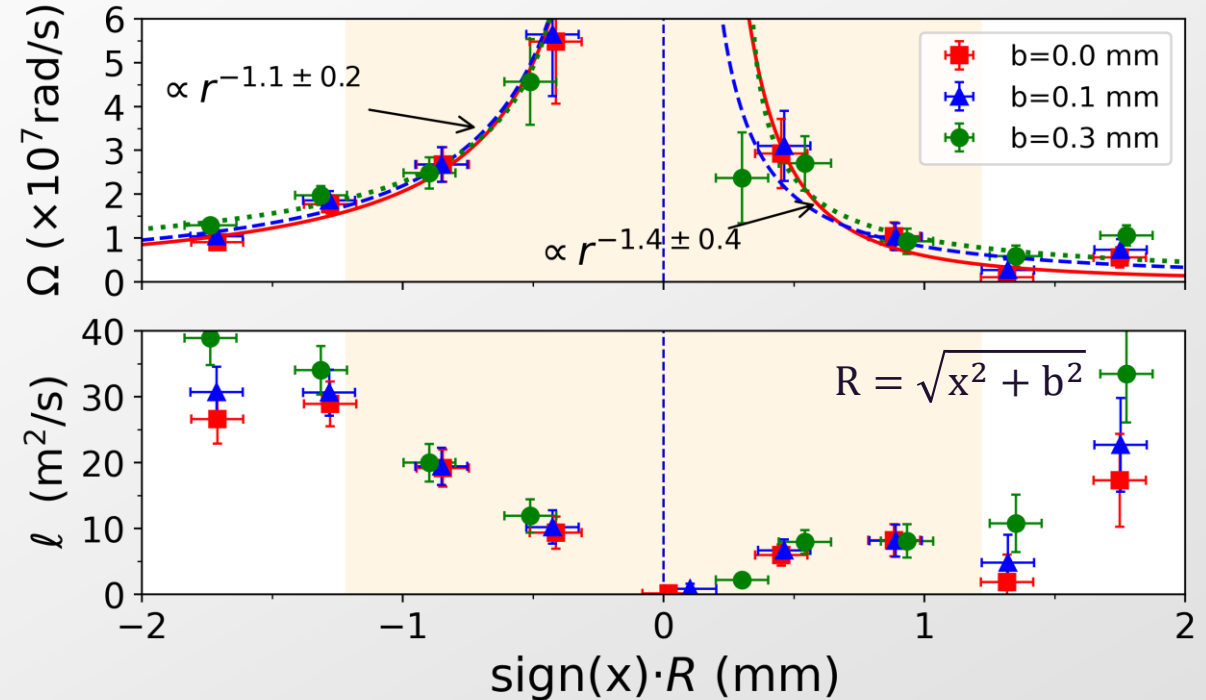


- Change in velocity component sign: **rotation**
- **Rotation velocity generally in range 15 – 20 km/s, as predicted by self-similar model**
- Maximum velocity ~ 25 km/s
- Velocity not proportional to radius: **differential rotation**
- **~ 1 to 2 full rotations** over duration of the experiment

The 2-D velocity field indicates that the flow stratification is quasi-Keplerian



[Valenzuela-Villaseca+ JPP (in prep.)]

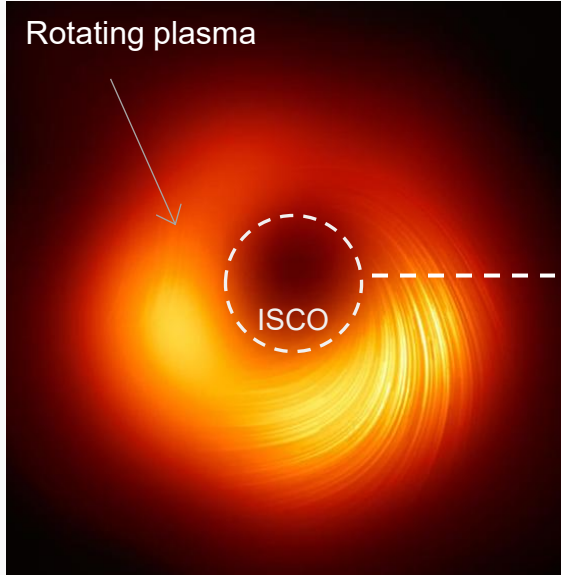


Rotation velocity stratification

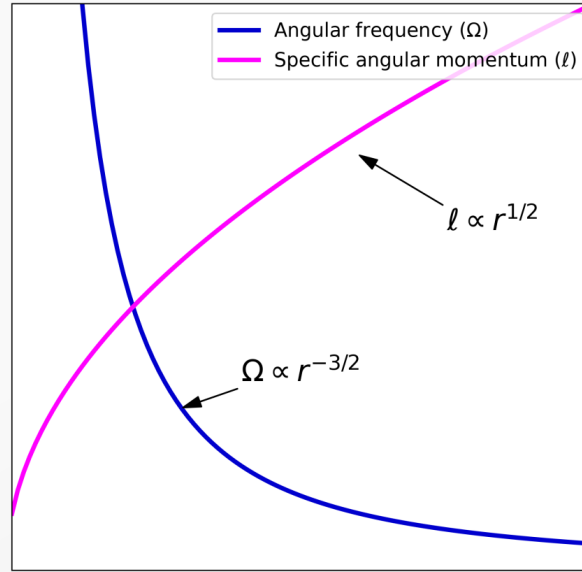
- Angular frequency (Ω) decreases with radius
- Angular momentum (ℓ) increases with radius

[Valenzuela-Villaseca+ PRL 2023]

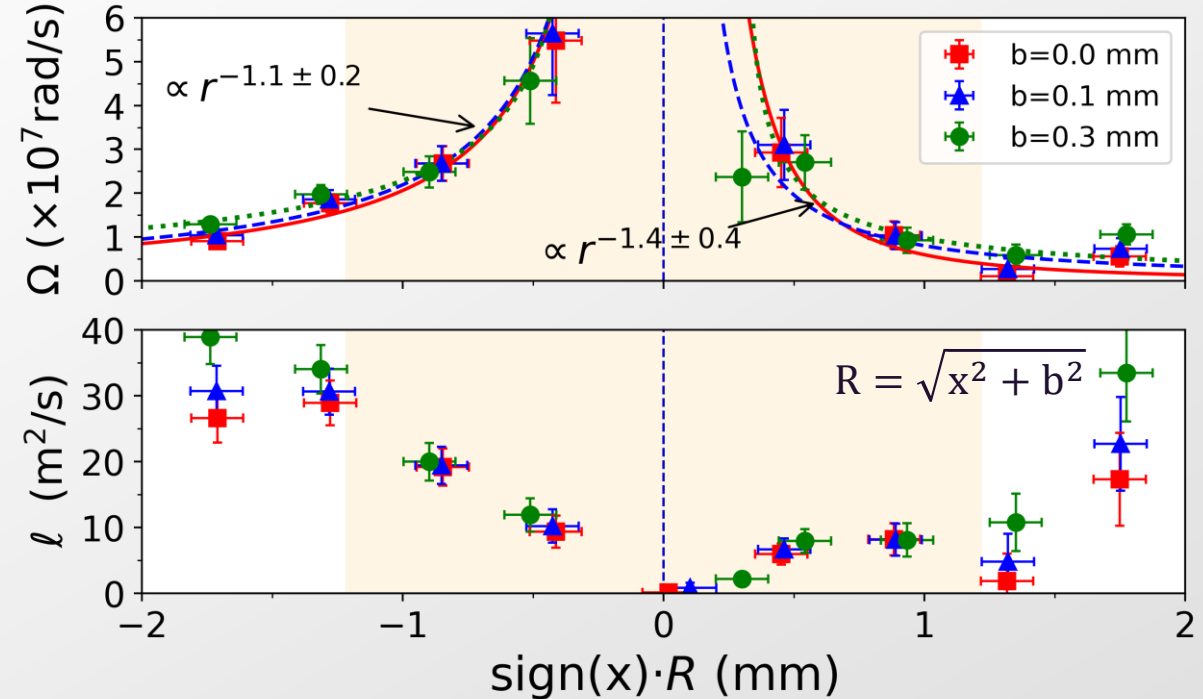
Stratified quasi-Keplerian flows capture key properties of accretion disks in the absence of actual gravity



[Akiyama+ (EHT) ApJ 2021]



Radial distance



Quasi-Keplerian rotation

- **Hydro stable** when $d\ell/dr > 0$ (Rayleigh criterion)
- MHD unstable under **Magneto-Rotational Instability (MRI)** when $d\Omega/dr < 0$
- Estimated **MRI growth time on RPX-like experiments** $\sim 100 \text{ ns}$ [Balbus & Hawley 1991]

Take-home message: the physics of rotating plasmas is a largely unexplored frontier by the HED community with fascinating scope

Summary

1. Pulsed-power experiments have the potential to address important issues in accretion disk astrophysics
2. RPX is a platform that drives rotating plasma flows and axial jets that transport angular momentum
3. Flow velocity field is quasi-Keplerian, mimicking the stability properties of gravity-driven accretion disks

In the (near) future...

- Destabilizing effects of axial magnetic fields
- Long-time evolution on new, long drive pulsed-power generators (e.g. Rama & Bertha at UCSD, PUFFIN at MIT)
- Laser-driven platforms at OMEGA **AngMomentumDisk** (PIs: Swadling/Suzuki-Vidal) and **MagOmegaEff** (PI: Valenzuela-Villaseca; first shot day 11 Jan 2024) will explore complementary regimes of magnetization and magnetic Prandtl number

